A FINITE PROJECTIVE MODULE MODULO AN IDEAL LIFTS AFTER AN ETALE RING EXTENSION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Cayley-Hamilton and the characteristic polynomial of an endomorphism of a finite free module. the author cites Algebra, Lemma algebra-lemma-charpoly at proof lines 2033-2035. Supplies the polynomial f whose roots control the idempotent.
- Lifting of idempotents along an etale ring map that is an isomorphism modulo the ideal. the statement counted by the sibling card S1-algebra-lift-idempotent at 1784-1803, cited at proof lines 2058-2060. Produces the idempotent of the finite extension ring that is used to define the lifted module.
- A finite projective module is a direct summand of a finite free module, so it is the image of an idempotent projector. used at proof lines 2028-2032. Supplies the projector whose lift produces the lifted module.

Source reference. Original lemma 07M5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9715–9715. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00DX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 514–514. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07M4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9714–9714. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a ring, let $I \subset A$ be an ideal. Let $\overline{P}$ be a finite projective A/I-module. Then there exists an étale ring map $A \to A'$ which induces an isomorphism $A/I \to A'/IA'$ and a finite projective A'-module P' lifting $\overline{P}$.*

Proof. We can choose an integer n and a direct sum decomposition $(A/I)^{\oplus n} = \overline{P} \oplus \overline{K}$ for some R/I-module $\overline{K}$. Choose a lift $\varphi : A^{\oplus n} \to A^{\oplus n}$ of the projector $\overline{p}$ associated to the direct summand $\overline{P}$. Let $f \in A[x]$ be the characteristic polynomial of φ. Set $B = A[x]/(f)$. By Cayley-Hamilton (Algebra, Lemma 00DX) there is a map $B \to \mathrm{End}_A(A^{\oplus n})$ mapping x to φ. For every prime $\mathfrak{p} \supset I$ the image of f in $\kappa(\mathfrak{p})$ is $(x-1)^r x^{n-r}$ where r is the dimension of $\overline{P} \otimes_{A/I} \kappa(\mathfrak{p})$. Hence $(x-1)^n x^n$ maps to zero in $B \otimes_A \kappa(\mathfrak{p})$ for all $\mathfrak{p} \supset I$. Thus $x(1-x)$ is contained in every prime ideal of B/IB. Hence $x^N(1-x)^N$ is contained in IB for some $N \geq 1$. It follows that $x^N + (1-x)^N$ is a unit in B/IB and that

$$\overline{e} = \text{image of } \frac{x^N}{x^N + (1-x)^N} \text{ in } B/IB$$

is an idempotent as both assertions hold in $\mathbf{Z}[x]/(x^N(x-1)^N)$. The image of $\overline{e}$ in $\mathrm{End}_{A/I}((A/I)^{\oplus n})$ is

$$\frac{\overline{p}^N}{\overline{p}^N + (1-\overline{p})^N} = \overline{p}$$

as $\overline{p}$ is an idempotent. After replacing A by an étale extension A' as in the lemma, we may assume there exists an idempotent $e \in B$ which maps to $\overline{e}$ in B/IB, see Lemma 07M4. Then the image of e under the map

$$B = A[x]/(f) \longrightarrow \mathrm{End}_A(A^{\oplus n}).$$

is an idempotent element p which lifts $\overline{p}$. Setting $P = \mathrm{Im}(p)$ we win. □